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CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS **DATE: 22/07/2026**

EXPERIMENT 8 -- Query with Sub Query& Correlated Query A MySQL subquery is query

AIM:

To perform **Subqueries using SELECT, INSERT, UPDATE, and DELETE commands** in the MySQL milk_dairys database.

PROCEDURE:

1. Open **MySQL** and create the database milk_dairys.
2. Select the database using:
3. USE milk_dairys;
4. Create the required tables such as milk_details and insert sample records.

   Use a **SELECT subquery** to retrieve records based on the result of another query.
5. Use an **INSERT subquery** to insert records obtained from another table using a subquery.
6. Use an **UPDATE subquery** to update records based on values returned by another query.
7. Use a **DELETE subquery** to delete records based on the result of another query.
8. Execute each query and verify the output.
9. Check that the required records are correctly selected, inserted, updated, and deleted.
10. Observe the results and confirm that the subqueries execute successfully..

PROGRAM:

1. CREATE TABLE

```
 CREATE TABLE milk_dairys (
name VARCHAR(20),
liters INT,
amount INT,
account_number BIGINT,
phone_number BIGINT,
bank_name VARCHAR(20)
);
```

THEN INSERT VALUES:

```
INSERT INTO milks_dairys(name, liters, amount, account_number, phone_number, bank_name)
```

VALUES
('Shanthi', 5, 320, 9679411723, 8639516519, 'SBI'),
('Nagu', 4, 290, 4192233411, 7182111462, 'SBI'),
('Sai', 7, 500, 497481234, 7670972570, 'SBI'),
('Venu', 9, 650, 727446123, 9204639364, 'Canara'),
('Sugana', 10, 700, 825849623, 6305857568, 'Andhra'),
('Sekar', 8, 600, 889241729, 9180091111, 'SBI'),
('Kamala', 15, 1500, 6284942417, 9132277911, 'SBI'),
('Ramu', 11, 950, 723967941, 8918286963, 'SBI'),
('Sita', 9, 650, 5492411763, 9350005229, 'Andhra'),
('Laxmi', 4, 290, 46418234748, 891991835, 'Canara');

OUTPUT:

```
mysql> SELECT*FROM milk_dairys;
+---------+--------+--------+----------------+--------------+-----------+
| name    | liters | amount | account_number | phone_number | bank_name |
+---------+--------+--------+----------------+--------------+-----------+
| Shanthi |      5 |    320 |     9679411723 |   8639516519 | SBI       |
| Nagu    |      4 |    290 |     4192233411 |   7182111462 | SBI       |
| Sai     |      7 |    500 |      497481234 |   7670972570 | SBI       |
| Venu    |      9 |    650 |      727446123 |   9204639364 | Canara    |
| Sugana  |     10 |    700 |      825849623 |   6305857568 | Andhra    |
| Sekar   |      8 |    600 |      889241729 |   9180091111 | SBI       |
| Kamala  |     15 |   1500 |     6284942417 |   9132277911 | SBI       |
| Ramu    |     11 |    950 |      723967941 |   8918286963 | SBI       |
| Sita    |      9 |    650 |     5492411763 |   9350005229 | Andhra    |
| Laxmi   |      4 |    290 |    46418234748 |    891991835 | Canara    |
+---------+--------+--------+----------------+--------------+-----------+
10 rows in set (0.00 sec)
```

1 SELECT AMOUNT GREATER THAN THE AVERAGE(amount)

SELECT name,amount
FROM milk_dairys
WHERE amount>(SELECT AVG(amount) FROM milk_dairys);

OUTPUT:

```
+--------+--------+
| name   | amount |
+--------+--------+
| Venu   |    650 |
| Sugana |    700 |
| Kamala |   1500 |
| Ramu   |    950 |
| Sita   |    650 |
+--------+--------+
5 rows in set (0.01 sec)
```

2. UPDATE AMOUNT WHERE MAX LITERS

UPDATE milk_dairys
SET amount=amount+100WHERE liters=(SELECT max_liters FROM(SELECT MAX(liters)max_liters FROM milk_dairys)AS temp);

OUTPUT:

```
mysql> SELECT*FROM milk_dairys;
+---------+--------+--------+----------------+--------------+-----------+
| name    | liters | amount | account_number | phone_number | bank_name |
+---------+--------+--------+----------------+--------------+-----------+
| Shanthi |      5 |    320 |     9679411723 |   8639516519 | SBI       |
| Nagu    |      4 |    290 |     4192233411 |   7182111462 | SBI       |
| Sai     |      7 |    500 |      497481234 |   7670972570 | SBI       |
| Venu    |      9 |    650 |      727446123 |   9204639364 | Canara    |
| Sugana  |     10 |    700 |      825849623 |   6305857568 | Andhra    |
| Sekar   |      8 |    600 |      889241729 |   9180091111 | SBI       |
| Kamala  |     15 |   1600 |     6284942417 |   9132277911 | SBI       |
| Ramu    |     11 |    950 |      723967941 |   8918286963 | SBI       |
| Sita    |      9 |    650 |     5492411763 |   9350005229 | Andhra    |
| Laxmi   |      4 |    290 |    46418234748 |    891991835 | Canara    |
+---------+--------+--------+----------------+--------------+-----------+
```

3. UPDATE AMOUNT WHERE MAX AMOUNT

UPDATE milk_dairys
SET amount=amount+200
WHERE amount=(SELECT max_amount FROM(SELECT MAX(amount) AS max_amount FROM milk_dairys) AS temp);

OUTPUT:

```
mysql> SELECT*FROM milk_dairys;
+---------+--------+--------+----------------+--------------+-----------+
| name    | liters | amount | account_number | phone_number | bank_name |
+---------+--------+--------+----------------+--------------+-----------+
| Shanthi |      5 |    320 |     9679411723 |   8639516519 | SBI       |
| Nagu    |      4 |    290 |     4192233411 |   7182111462 | SBI       |
| Sai     |      7 |    500 |      497481234 |   7670972570 | SBI       |
| Venu    |      9 |    650 |      727446123 |   9204639364 | Canara    |
| Sugana  |     10 |    700 |      825849623 |   6305857568 | Andhra    |
| Sekar   |      8 |    600 |      889241729 |   9180091111 | SBI       |
| Kamala  |     15 |   1800 |     6284942417 |   9132277911 | SBI       |
| Ramu    |     11 |    950 |      723967941 |   8918286963 | SBI       |
| Sita    |      9 |    650 |     5492411763 |   9350005229 | Andhra    |
| Laxmi   |      4 |    290 |    46418234748 |    891991835 | Canara    |
+---------+--------+--------+----------------+--------------+-----------+
10 rows in set (0.00 sec)
```

4. UPDATE AMOUNT WHERE AVG_LITERS

UPDATE milk_dairys
SET amount=amount+500
WHERE liters>(SELECT avg_liters FROM(SELECT AVG(liters) AS avg_liters FROM milk_dairys) AS temp);

OUTPUT:

mysql> SELECT*FROM milk_dairys;

name	liters	amount	account_number	phone_number	bank_name
Shanthi	5	320	9679411723	8639516519	SBI
Nagu	4	290	4192233411	7182111462	SBI
Sai	7	500	497481234	7670972570	SBI
Venu	9	1150	727446123	9204639364	Canara
Sugana	10	1200	825849623	6305857568	Andhra
Sekar	8	600	889241729	9180091111	SBI
Kamala	15	2300	6284942417	9132277911	SBI
Ramu	11	1450	723967941	8918286963	SBI
Sita	9	1150	5492411763	9350005229	Andhra
Laxmi	4	290	46418234748	891991835	Canara

10 rows in set (0.00 sec)

5)INSERT INTO milk_backup(name ,liters,amount)

INSERT INTO milk_backup (name, litres, amount)
SELECT name, litres, amount
FROM milk_dairys
WHERE litres = (
SELECT max_litres
FROM (
SELECT MAX(litres) AS max_litres
FROM milk_dairys
) AS temp
);

OUTPUT:

mysql> SELECT*FROM milk_backup;

name	liters	amount	account_number	phone_number	bank_name
Shanthi	5	320	9679411723	8639516519	SBI
Nagu	4	290	4192233411	7182111462	SBI
Sai	7	500	497481234	7670972570	SBI
Venu	9	650	727446123	9204639364	Canara
Sugana	10	700	825849623	6305857568	Andhra
Sekar	8	600	889241729	9180091111	SBI
Kamala	15	1500	6284942417	9132277911	SBI
Ramu	11	950	723967941	8918286963	SBI
Sita	9	650	5492411763	9350005229	Andhra
Laxmi	4	290	46418234748	891991835	Canara
Kamala	15	2300	NULL	NULL	NULL

11 rows in set (0.00 sec)

6)DELETE MIN_LITERS

DELETE FROM milk_dairys
WHERE liters=(SELECT min_liters FROM(SELECT MIN(liters) AS min_liters FROM milk_dairys)AS temp);

OUTPUT:

```
mysql> SELECT*FROM milk_dairys;;
+---------+--------+--------+----------------+--------------+-----------+
| name    | liters | amount | account_number | phone_number | bank_name |
+---------+--------+--------+----------------+--------------+-----------+
| Shanthi |      5 |    320 |     9679411723 |   8639516519 | SBI       |
| Sai     |      7 |    500 |      497481234 |   7670972570 | SBI       |
| Venu    |      9 |   1150 |      727446123 |   9204639364 | Canara    |
| Sugana  |     10 |   1200 |      825849623 |   6305857568 | Andhra    |
| Sekar   |      8 |    600 |      889241729 |   9180091111 | SBI       |
| Kamala  |     15 |   2300 |     6284942417 |   9132277911 | SBI       |
| Ramu    |     11 |   1450 |      723967941 |   8918286963 | SBI       |
| Sita    |      9 |   1150 |     5492411763 |   9350005229 | Andhra    |
+---------+--------+--------+----------------+--------------+-----------+
8 rows in set (0.00 sec)
```

RESULT

Thus, the Subqueries using SELECT, INSERT, UPDATE, and DELETE commands were successfully executed in the milk_dairys database. The required records were successfully retrieved, inserted, updated, and deleted using nested queries, and the results were verified successfully.